

TRACEMAIL AI

FORENSIC INVESTIGATION REPORT

From Inbox to Attack Map.

CASE ID TM-2026-00142	SEVERITY CRITICAL	RISK SCORE 94/100
STATUS Open • Investigation	THREAT Credential Phishing	MODE Demo Environment

Executive Finding

Assessment

The message exhibits multiple independent phishing indicators: authentication/header anomalies, sender and Reply-To inconsistency, a lookalike domain, a suspicious URL pattern, and social-engineering language. The combined evidence supports a **high-confidence malicious classification**.

94	81%	3	2
RISK SCORE	GEO CONFIDENCE	MATCHED RECIPIENTS	KNOWN IOCs

Analyst note: Infrastructure geolocation identifies observable network infrastructure. It does **not** establish the physical location or identity of the attacker.

1. Email & Header Forensics

Field	Observed value	Evidence class
Subject	Urgent: Verify your account immediately	OBSERVED
From	security-alert@microsoft-support.example	OBSERVED
Reply-To	verify-account@secure-check.example	OBSERVED
Authentication	SPF: FAIL • DKIM: FAIL • DMARC: FAIL	OBSERVED
URL	https://secure-check.example/verify	OBSERVED
Attachment	None	OBSERVED

2. Explainable Risk Breakdown

Signal	Impact	Result
Authentication failure	Strong malicious indicator	HIGH
Lookalike sender/domain	Brand impersonation risk	HIGH
Reply-To mismatch	Potential credential redirection	HIGH
Suspicious URL structure	Credential collection risk	HIGH
Urgency / social engineering	Pressure to act quickly	MEDIUM
Infrastructure reputation	Suspicious observable infrastructure	HIGH

AI conclusion

High-confidence phishing. Multiple independent signals converge on credential theft / account takeover intent. Classification is grounded in extracted email evidence and enriched intelligence; no claim is made about user interaction or successful compromise.

3. Infrastructure Intelligence

Node	Observable intelligence	Confidence
Recipient	India • Mailbox	OBSERVED
Mail relay	Singapore • ASN / relay infrastructure	ENRICHED
Suspicious infrastructure	United States • Hosting / proxy indicator	ENRICHED
Physical attacker location	Unknown	UNKNOWN

Geolocation limitation

IP geolocation can point to a VPN, proxy, cloud host, mail relay, compromised server, or other infrastructure. Therefore the report describes **observable infrastructure**, not the attacker's physical location.

4. Campaign Correlation & Attack DNA

Fingerprint component	Value
Campaign DNA	PHISH-BRAND-IMPERSONATION-042
Similarity to prior campaign	87%
Mutation detected	Yes — infrastructure/domain variation
Historical reuse	Related infrastructure observed in prior demo cases
Blast radius	3 matching recipients / 4 matching messages

5. Forensic Timeline

Time	Event	Class
09:41	Message received by monitored mailbox	OBSERVED
09:41	Header authentication checks evaluated	OBSERVED
09:42	IOC extraction completed	OBSERVED
09:42	Threat model classified message as high risk	INFERRED
09:43	Infrastructure intelligence enriched	ENRICHED
09:43	Related campaign fingerprint matched	INFERRED
09:44	Analyst review / response decision pending	OBSERVED

6. Evidence Gaps

Known unknowns

- Physical attacker location: **UNKNOWN**
- User clicked the link: **UNKNOWN / not observed**
- Credentials were entered: **UNKNOWN / not observed**
- Successful account compromise: **UNKNOWN**
- Attribution to a specific person: **NOT ESTABLISHED**

7. Recommended Response

Priority	Action	Authorization
1	Quarantine the message	Human approval required
2	Block confirmed malicious IOC(s)	Human approval required
3	Hunt for matching sender/domain/URL across mailboxes	Analyst action
4	Review affected accounts and authentication logs	Analyst action
5	Preserve original email + headers as evidence	Analyst action

Chain-of-evidence note

This report distinguishes directly observed email artifacts from external enrichment and AI inference. Suspicious content is treated as untrusted input; links and attachments are not executed as part of this demo workflow.